7(a)(i)	$I = \frac{P}{S} \Rightarrow P = IS$ $P = 5.0 \times 10^2 \left(2.6 \times 10^{14}\right)$	C1
	$= 1.3 \times 10^{17}$ W	A1
(a)(ii)	1.3×10^{17} W	A1
(a)(iii)	$I = \frac{P}{S} = \frac{P}{4\pi r^2}$	M1
	$= \frac{1.3 \times 10^{17}}{4\pi\left(6400 \times 10^3\right)^2}$	M1
	$= 253$ W m^{-2}	A0
(b)(i)	x-axis marked with an arrow labelled M at 900-1100 nm	B1
(b)(ii)	$\lambda_{max} T = k$ $1000\left(2900\right) = k$	C1
	$k = 2.9 \times 10^6$ nm K	A1
(b)(iii)	λ_{max} T = 2.90 x 10^6 nm K	
	Sun: λ_{max} = 2.90 x 10^6 / 5800 = 500 nm	A1
	Earth: λ_{max} = 2.90 x 10^6 / 290 = 10000 nm	A1
(b)(iv)	Sun's curve always above the original curve and peak at 500 nm	B1
(c)(i)	Radiation window – range of wavelengths which are transmitted/not absorbed by CO_2	B1
(c)(ii)	At 11000 nm, CO_2 absorbs close to 100% of radiation, so it is not a radiation window	B1
(d)	Sun's peak intensity in the 500 nm visible range. Energy radiated by Sun is able to pass through atmosphere as it is within the radiation window.	B1
	The visible light that reaches Earth is absorbed and re-radiated.	B1
	Earth's peak intensity is in the 10000 nm infrared range. Energy radiated by Earth is trapped by carbon dioxide as it is not in radiation window.	B1
	This leads to increase in temperature within the atmosphere.	

(e)(i)	$m = \rho V \Rightarrow v = \dfrac{m}{\rho}$	C1
	$v = \dfrac{2.8 \times 10^{18}}{1.0 \times 10^{3}} = 2.8 \times 10^{15}\ \text{m}^{-3}$	A1
(e)(ii)	Ice floats thus upthrust = weight of ice. And upthrust equals weight of displaced sea water. When ice melts, melted ice water only occupies the same volume as submerged part of ice pack	M1
	Hence no change to sea level.	A1